

Module-2 Conformal and Bilinear Transformations

Transformation: For every point (x, y) in the z -plane, the relation $w = f(z)$ defines the corresponding point (u, v) in the w -plane.

i.e., Transformation of z -plane into w -plane.

Ex $x=1, x=3, y=0, y=3$

$$w = z + (2+i)$$

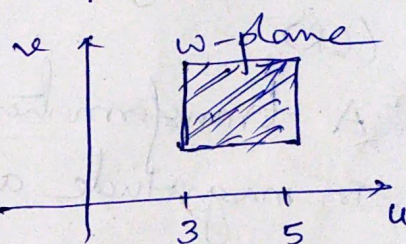
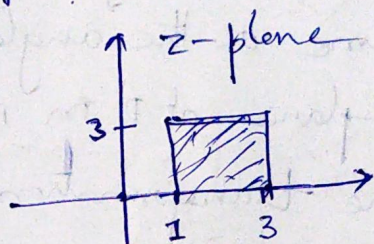
$$= x+iy + 2+i = x+2 + i(y+1) = u+iv$$

$$x=1, u=3;$$

$$x=3, u=5;$$

$$y=0, v=1$$

$$y=3, v=4$$



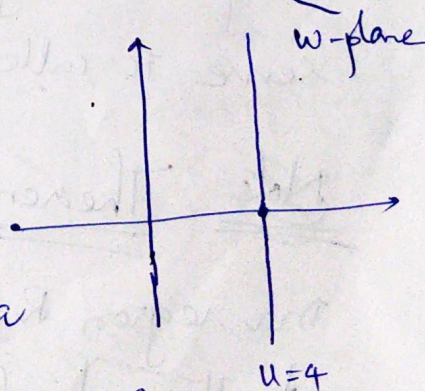
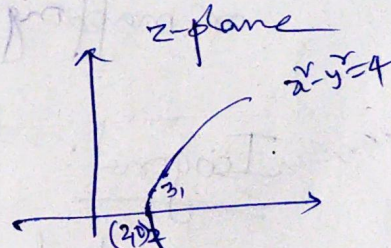
Ex $x^2 - y^2 = 4, w = z^2$

$$w = z^2 = (x+iy)^2 = x^2 - y^2 + i2xy$$

$$u = x^2 - y^2 = 4, v = 2xy$$

$$(1,1), (2,3), (3,5).$$

$$\hookrightarrow u=0, \hookrightarrow u=$$



$\therefore$ Image of the curve $x^2 - y^2 = 4$ is a straight line $u=4$ parallel to v -axis in w -plane.

Conformal Transformation:

Def. let two curves C_1, C_2 in the z -plane intersect at the point P and the corresponding curve C'_1, C'_2 in the w -plane intersect at P' . If the angle of intersection of the curves at P in z -plane is the same as the angle of intersection of the curves of w -plane at P' in magnitude and sense, then the transformation is called conformal.

(OR)

A transformation or mapping that preserves both in magnitude and sense, the transformation or mapping is said to be conformal mapping.

Isogonal: A mapping that preserves the magnitudes of angles but not necessarily the sense is called isogonal.

Note: Theorem. If $f(z)$ is analytic and $f'(z) \neq 0$ in a region R , then the mapping $w=f(z)$ is conformal at all points of R .

Note: i) An analytic function is conformal at z_0 if $f'(z_0) \neq 0$.

ii) Points where $f'(z) = 0$ are called critical points. Also the points where $f'(z) \neq 0$ are called ordinary points.

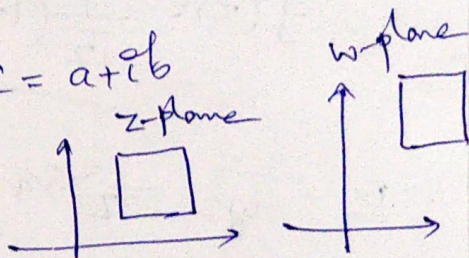
Standard Transformations:

1. Translation $w = z + c$, $c = a + ib$

$$\therefore u + iv = x + iy + a + ib$$

$$\therefore u = x + a, \quad v = y + b$$

$$x = u - a, \quad y = v - b$$



2. Rotation: $w = ze^{i\theta}$

consider $w = ze^{i\pi/4}$

$$\begin{aligned} u + iv &= (x + iy)(\cos \pi/4 + i \sin \pi/4) \\ &= (x + iy)\left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}\right) = (x + iy)\left(\frac{1+i}{\sqrt{2}}\right) \\ &= \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}} + i\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right) \\ &= \frac{1}{\sqrt{2}}((x-y) + i(x+y)) \end{aligned}$$

$$u = \frac{1}{\sqrt{2}}(x-y); \quad v = \frac{1}{\sqrt{2}}(x+y)$$

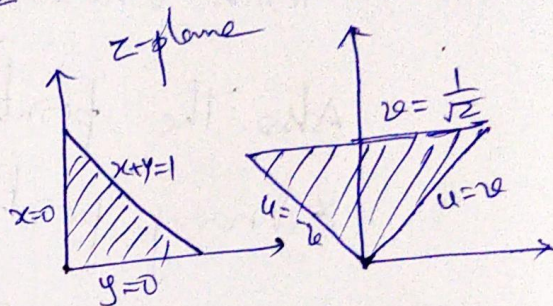
i) $x=0, \quad u = -\frac{1}{\sqrt{2}}y; \quad v = \frac{1}{\sqrt{2}}y \Rightarrow v = -u$

ii) $y=0; \quad u = \frac{x}{\sqrt{2}}, \quad v = \frac{x}{\sqrt{2}} \Rightarrow v = u$

iii) $x+y=1, \quad v = \frac{1}{\sqrt{2}}$

$$f(z) = \frac{1}{\sqrt{2}}[(x-y) + i(x+y)]$$

$$f'(z) = \frac{1}{\sqrt{2}}(1+i)$$



Ex $w = \sqrt{2}e^{i\pi/4} \Rightarrow u+iv = (x+iy)\sqrt{2}(\cos \frac{\pi}{4} + i\sin \frac{\pi}{4})$
 $= (x+iy)(1+i) = x-y + i(x+y)$

$$\therefore u = x-y; \quad v = x+y$$

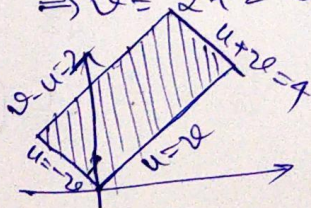
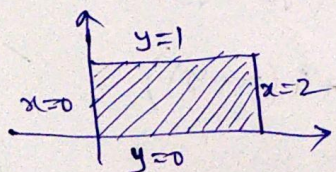
i) $x=0, \quad u = -y, \quad v = y \Rightarrow v = -u$

ii) $y=0, \quad u = x, \quad v = x \Rightarrow u = v$

iii) $x+y=1; \quad u = x-1, \quad v = x+1 \Rightarrow x = v-1$
 $\Rightarrow u = v-1-1 \Rightarrow u-v = -2$

iv) $x=2; \quad u = 2-y, \quad v = 2+y \Rightarrow y = 2-u$

$$\Rightarrow v = 2+2-u \Rightarrow u+v = 4$$



3. magnification: $w = cz$ where c is a real quantity.

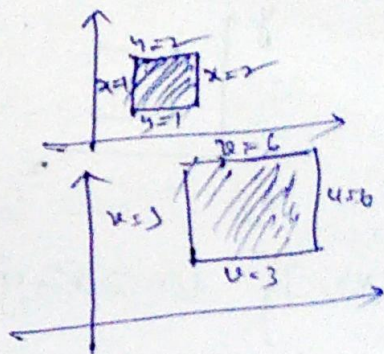
- i) The figure in w -plane is magnified c -times the size of the figure in z -plane.
- ii) Both figures in z -plane & w -plane are similar

Ex ~~$w = 3z$~~ ,
Determine the region in w -plane on the transformation of rectangular region enclosed by $x=1, y=1, x=2$ & $y=2$ in the z -plane, the transformation is $w = 3z$. z -plane

$$w = 3z$$

$$w = 3(x + iy) = 3x + i3y$$

$$u = 3x; v = 3y$$



Magnification and Rotation: $w = cz$

where c, z are complex numbers.

$$c = ae^{i\alpha}, z = re^{i\theta}, w = Re^{i\phi}$$

$$\Rightarrow w = cz$$

$$Re^{i\phi} = ae^{i\alpha} re^{i\theta} = are^{i(\alpha+\theta)}$$

$$\therefore R = ar; \phi = \alpha + \theta$$

$\therefore w = cz$ corresponds to a rotation, together with magnification.

$$w = cz \text{ (so) } u+iv = (a+ib)(x+iy)$$

$$u+iv = ax-by+i(ay+bx)$$

$$\therefore u = ax-by; \quad v = ay+bx$$

$$u = ax-by \text{ --- (1)} \quad \Rightarrow \quad au = a^2x^2 - aby$$

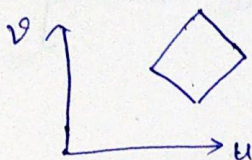
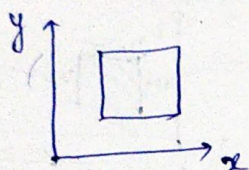
$$v = ay+bx \text{ --- (2)}$$

$$bv = b^2x^2 + aby$$

$$\underline{x^2(a^2+b^2) = au+bv}$$

$$x = \frac{au+bv}{a^2+b^2};$$

$$y = \frac{-bu+av}{a^2+b^2}$$



(*) If $u = 2x^2+y^2$ and $v = \frac{y^2}{x}$, shows that the curves $u = \text{constant}$ and $v = \text{constant}$ cut orthogonally at all intersections but that the transformation $w = u+iv$ is not conformal.

$$u = 2x^2+y^2, \quad v = \frac{y^2}{x}$$

$$u = \text{constant} \Rightarrow 2x^2+y^2 = C; \quad v = \text{constant} \Rightarrow \frac{y^2}{x} = C'$$

by differentiating $4x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2x}{y}$

$$v = \text{constant} \Rightarrow \frac{y^2}{x} = C' \Rightarrow y^2 = xC' \Rightarrow 2y \frac{dy}{dx} = \frac{C'}{x}$$

$$\frac{dy}{dx} = \frac{C'}{2y} = \frac{y^2}{x} \cdot \frac{1}{2y} = \frac{y}{2x} \quad \left(\because C' = \frac{y^2}{x} \right)$$

$$\therefore m_1 m_2 = -\frac{2x}{y} \cdot \frac{y}{2x} = -1$$

$\therefore$ Two curves cut orthogonally.

$$\frac{\partial u}{\partial x} = 4x; \quad \frac{\partial u}{\partial y} = 2y; \quad \frac{\partial v}{\partial x} = -\frac{y^2}{x^2}; \quad \frac{\partial v}{\partial y} = \frac{2y}{x}$$

$\therefore$ The Cauchy-Riemann equations are not satisfied.

$\therefore f(z) = u + iv$ is not analytic $\Rightarrow f(z)$ is not conformal.

Ex For the conformal transformation $w = z^2$,

i) The coefficient of magnification at $z = 2+i$ is $2\sqrt{5}$

$$\begin{array}{l|l} w = f(z) = z^2, & f'(z) = 2z \\ f'(2+i) = 2(2+i) = 4+2i & \text{at } z = 1+i, f'(z) = 2(1+i) = 2+2i \\ |f'(2+i)| = \sqrt{16+4} = 2\sqrt{5} & |f'(1+i)| = \sqrt{4+4} = 2\sqrt{2} \end{array}$$

ii) The angle of rotation at $z = 2+i$ is $\tan^{-1} 0.5$

$$f'(2+i) = 4+2i, \quad \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2}{4}\right) = \underline{\underline{\tan^{-1} 0.5}}$$

$$z = 1+i \quad \therefore f'(1+i) = 2(1+i) = 2+2i$$

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1} 2 = \underline{\underline{\frac{\pi}{4}}}$$